Eyub Celebioglu

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Research Interests

Machine learning, reinforcement learning, computer vision and robotics. I plan to pursue a PhD.

Education

Université de Lorraine · *co-accredited with CentraleSupélec* 2026 – Present
MSc in Computer Science — AI, Vision & Robotics track (IA2VR), second year *Nancy, France*

- Coursework: deep learning, image processing, planning/learning/control for robotics, geometric models for vision, collective robotics.

Université de Lorraine · *co-accredited with CentraleSupélec* 2025 – 2026
MSc in Computer Science, first year — AI-oriented electives *Nancy, France*

- Selected courses: artificial intelligence, computer vision, autonomous mobile robotics, speech processing, neuromorphic computing, data parallelism; research project.

ISIMA – Université Clermont Auvergne 2022 – 2025
BSc in Computer Science *Clermont-Ferrand, France*

Université Clermont Auvergne 2021 – 2022
First year of BSc in Physics, then transferred to Computer Science *Clermont-Ferrand, France*

Research Experience

Autonomous Navigation in Uncertain Environments 2025 – 2026 (6 months)
First-year Master's research project, supervised by Olivier Buffet (Inria / LORIA) · team of 3 *Nancy, France*

- Developed the 2D robot simulator used by the team: kinematic model, collision checking with inflated obstacles, 5-ray LIDAR.
- Designed and trained the DQN controller in PyTorch (MLP 13-128-128-5, experience replay, target network) that follows RRT waypoints while avoiding obstacles; 3000 episodes on 5 maps with random starts, 73–91% success per map.
- Iterated over three training versions: diagnosed reward hacking (agent spinning in place) and miscalibrated wall penalties; oversampled rare collision transitions ×10 in the replay buffer.
- The team also compared RRT, RRT* and **RRTSC**, a shortcut variant we proposed (≈70% fewer path nodes than RRT, at a fraction of RRT* runtime).

Selected Projects

Neural Networks from Scratch | *C++* code

- Multi-layer perceptron and backpropagation in plain C++ without any ML library, trained to learn XOR.

Deep Q-Learning for Snake | *Python, reinforcement learning* code

- Trained a DQN agent to play Snake from the game state; precursor to the navigation work.

Anime Face Generation with a DCGAN | *Python, GAN, Kaggle GPU* code

- Trained a deep convolutional generator and discriminator on GPU to generate anime character faces.

Local LLM Voice Assistant | *Python, Ollama, Docker* code

- Served open-source LLMs (Llama) locally in Docker containers, with a voice interface and speech synthesis.

3D Physics and Rendering Engine | *C++, OpenGL* code

- Modular engine with a custom .obj parser and real-time collisions; extended into a voxel engine with procedural terrain.

Technical Skills

Programming: Python, C/C++, Java, C#, SQL, R, Prolog
Machine learning: PyTorch, NumPy, Matplotlib; reinforcement learning (DQN), GANs
Robotics & systems: ROS, motion planning (RRT, RRT*), OpenMP, OpenGL, Docker, Linux, Git, CPLEX, SageMath
Languages: French (native), English (B2)